

Introduction to Quantum Computing — Exam Cheatsheet

RWTH Aachen (Unruh, SS 2025) — Final: 20 Aug 2026 — ordered by marks in the practice e-tests

The 30-second core

- Prob. distribution: $d \in \mathbb{R}^N$, $d_i \geq 0$, $\sum d_i = 1$.
- Stochastic matrix: *every column* is a distribution.
- Quantum state: $\psi \in \mathbb{C}^N$, $\sum |\psi_i|^2 = 1$.
- Unitary: $U^\dagger U = UU^\dagger = I$ († = transpose *and* conjugate).
- Measure: $\Pr[x] = |\psi_x|^2$, p.m.s. = e_x (state *changes*).
- Partial measure: sum $|\psi_i|^2$ over the alternative, zero out the rest, divide by $\sqrt{\Pr}$.
- Compose: $\otimes$ for states and matrices; $(A \otimes B)(\psi \otimes \phi) = A\psi \otimes B\phi$.
- Global phase c , $|c| = 1$: physically invisible. Say so, don't panic.

1. Tensor Products [74.5 marks in practice e-tests]

Definition

For systems A (possibilities $x_1 \dots x_N$) and B ($y_1 \dots y_M$), the composite AB has NM possibilities ordered $x_1 y_1, x_1 y_2, \dots, x_1 y_M, x_2 y_1, \dots, x_N y_M$, and state

$$\mu_{AB} = \mu_A \otimes \mu_B = \begin{pmatrix} (\mu_A)_1 \mu_B \\ \vdots \\ (\mu_A)_N \mu_B \end{pmatrix}$$

$$(\mu_{AB})_{x_i y_j} = (\mu_A)_{x_i} (\mu_B)_{y_j}.$$

For matrices (S is $N \times N$, T is $M \times M$):

$$S \otimes T = \begin{pmatrix} S_{11}T & \cdots & S_{1N}T \\ \vdots & \ddots & \vdots \\ S_{N1}T & \cdots & S_{NN}T \end{pmatrix} \in \mathbb{C}^{NM \times NM}.$$

Apply S to A and T to $B \Rightarrow$ apply $S \otimes T$ to AB .

Algebra rules (usable without proof)

$$(A \otimes B)(\psi \otimes \phi) = A\psi \otimes B\phi, \quad (A \otimes B)(C \otimes D) = AC \otimes BD,$$

$$\psi \otimes (\phi + \chi) = \psi \otimes \phi + \psi \otimes \chi, \quad (\psi + \phi) \otimes \chi = \psi \otimes \chi + \phi \otimes \chi,$$

$$A \otimes (B + C) = A \otimes B + A \otimes C, \quad (A + B) \otimes C = A \otimes C + B \otimes C,$$

$$c\phi \otimes \psi = \phi \otimes c\psi = c(\phi \otimes \psi), \quad cA \otimes B = A \otimes cB = c(A \otimes B).$$

$\otimes$ is *not* commutative. It *is* associative and bilinear.

Entanglement

ψ_{AB} is **entangled** $\iff$ it *cannot* be written as $\psi_A \otimes \psi_B$. Quick 2-qubit test on $(a, b, c, d)^T$ (order 00, 01, 10, 11): product state $\iff ad - bc = 0$.

- $\beta_{00} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: $ad - bc = \frac{1}{2} - 0 \neq 0 \Rightarrow$ entangled.
- $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes |+\rangle$: $ad - bc = 0 \Rightarrow$ product.

Ex. $\mu_A = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)^T$, $\mu_B = \left(\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right)^T$.

$$\mu_A \otimes \mu_B = \frac{1}{\sqrt{2}} \left(\frac{1}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right)^T$$

with possibilities ordered 1a, 1b, 1c, 2a, 2b, 2c.

Dimension / counting facts

$x \in \mathbb{R}^n$, $y \in \mathbb{R}^m \Rightarrow x \otimes y \in \mathbb{R}^k$ with $k = nm$. n copies of a 2-vector $\Rightarrow 2^n$ entries. $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^{\otimes n}$: 2^n entries, every entry (so also the first) $= \left(\frac{1}{\sqrt{2}}\right)^n = 2^{-n/2}$.

Computing one, fast

Ex. $\begin{pmatrix} 1 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 2 \\ 3 \end{pmatrix} = (2, 3, 2, 3)^T$; $\begin{pmatrix} 5 \\ 3 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 4 \end{pmatrix} = (5, 20, 3, 12)^T$; $\left(\frac{1}{3}, \frac{2}{3}, \frac{1}{8}\right)^T \otimes (2, 1)^T = \left(\frac{2}{3}, \frac{1}{3}, \frac{4}{3}, \frac{2}{3}, \frac{1}{4}, \frac{1}{8}\right)^T$.

Ex. $\begin{pmatrix} 5 & 8 \\ -3 & 2 \end{pmatrix} \otimes \begin{pmatrix} 2 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 10 & 15 & 16 & 24 \\ 5 & -10 & 8 & -16 \\ -6 & -9 & 4 & 6 \\ -3 & 6 & 2 & -4 \end{pmatrix}$.

A 1×1 matrix $[c]$ tensored with M is just cM .

Tracing a circuit (the $\psi_1, \psi_2, \dots$ format)

1. Write the input as one big tensor product.
2. At each time slice, pad every idle wire with I : a gate on wire 2 of 3 is $I \otimes G \otimes I$.
3. Apply slice by slice, in ket notation (translate each ket individually — far less error-prone than multiplying 8×8 matrices).
4. Label and *keep* every intermediate $|\psi_i\rangle$; the marks are per state.

Ex. $(H \otimes H)|00\rangle = |+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$; then CNOT $\rightarrow \frac{1}{2}(|00\rangle + |01\rangle + |11\rangle + |10\rangle)$ — unchanged (it permutes the four equal terms).

Trap: order. $\psi_A \otimes \psi_B \neq \psi_B \otimes \psi_A$. On a 3-factor product, write the factors in the order the question gives them — reversing the order was a confirmed lost-marks error. Read off the wire/register order before you multiply anything.

Trap: block structure. In $S \otimes T$, the *outer* matrix supplies the scalars, the *inner* matrix is copied into every block. $S \otimes T$ has T -sized blocks, not S -sized ones.

Trap: $|+\rangle^{\otimes 3} \neq$ encoding. Encoding $|+\rangle$ in a repetition code is $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$, *not* $|+\rangle \otimes |+\rangle \otimes |+\rangle$.

2. Hamiltonian Dynamics [58.5 marks in practice e-tests]

The four facts

1. **Hamiltonian** $H \in \mathbb{C}^{N \times N}$ is always **Hermitian**: $H^\dagger = H$.
2. **Total energy** of state ψ : $E_{\text{total}} = \langle \psi | H | \psi \rangle$. For diagonal $H = \text{diag}(E_0, \dots, E_N)$ this equals $\sum_i |\alpha_i|^2 E_i$.
3. **Energy eigenstate** ($H\psi(0) = E\psi(0)$) evolves as $\psi(t) = e^{-iEt/\hbar} \psi(0)$.
4. **General state**: $\psi(t) = U_t \psi(0)$ with $U_t = e^{-iHt/\hbar}$. Waiting time t = applying U_t .

Notes set $\hbar = 1$. The notes write the Schrödinger equation as $\frac{d\psi(t)}{dt} = \frac{i}{\hbar} H(t) \psi(t)$; **all solved examples use** $e^{-iEt/\hbar}$

and $U_t = e^{-iHt}$ — use those signs when computing.

Matrix exponential — the procedure

For diagonal D : $e^D = \text{diag}(e^{D_{11}}, \dots, e^{D_{nn}})$. For diagonalizable $A = U^\dagger D U$: $e^A = U^\dagger e^D U$.

1. If H is already diagonal: exponentiate the entries, done.
 $U_t = \text{diag}(e^{-iE_{0t}}, \dots)$.
2. Otherwise: eigenvalues from $\det(H - \lambda I) = 0$.
3. For each λ : solve $(H - \lambda I)v = 0$, normalise v .
4. Write $\psi(0)$ in the eigenbasis, evolve each component by its own $e^{-iE_k t}$, sum back.

Eigen-decomposition drill (2×2)

$\det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} = \lambda^2 - (a+d)\lambda + (ad-bc) = 0$. A 2×2 matrix has **two** eigenvalues (possibly equal) — never stop after one.

Ex. $H = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$.

$\lambda = +1$: $(X - I)v = 0 \Rightarrow \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0 \Rightarrow v_1 = v_2 \Rightarrow |+\rangle$.

$\lambda = -1$: $(X + I)v = 0 \Rightarrow v_1 = -v_2 \Rightarrow |-\rangle$.

Evolve $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$: $\psi(t) = \frac{1}{\sqrt{2}}(e^{-it} |+\rangle + e^{+it} |-\rangle) = \cos t |0\rangle - i \sin t |1\rangle$.

Keep at hand: $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$.

Worked exam pattern (single-qubit computer)

Energy levels $\pi, 2\pi \Rightarrow H = \text{diag}(\pi, 2\pi)$, $U_t = \begin{pmatrix} e^{-i\pi t} & 0 \\ 0 & e^{-2i\pi t} \end{pmatrix}$. At $t = 1$: $U_1 = \text{diag}(-1, 1) = -Z \equiv Z$ (global phase). Driving the system adds δH ; if $\delta H \gg H$

then $U'_t \approx e^{-i\delta H t}$. With $\delta H = \begin{pmatrix} 0 & 1000 \\ 1000 & 0 \end{pmatrix}$ and $t = \frac{\pi}{2000}$:

$U \approx \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} = -iX \equiv X$.

Trap: sign of $H - \lambda I$ for negative λ . Subtracting $\lambda = -3$ means *adding* 3 on the diagonal.

Trap: trivial solution. $v = 0$ always solves $(H - \lambda I)v = 0$ and is never the eigenvector. You want the non-trivial null-space direction, then normalise.

Trap: Hermitian $\neq$ symmetric. $H^\dagger = H$ means conjugate and transpose: off-diagonals must be complex conjugates ($H_{12} = \overline{H_{21}}$), diagonal entries must be *real*.

3. Unitary Matrices [54.5 marks in practice e-tests]

Definition and test

U is unitary $\iff U^\dagger U = U U^\dagger = I$, where $U^\dagger = \overline{U}^T$ (transpose and complex-conjugate every entry).

Equivalent tests: columns are orthonormal; rows are orthonormal; U preserves $\|\cdot\|$ (so it maps quantum states to quantum states); $|\det U| = 1$.

Procedure to answer "is this unitary?": write $U^\dagger$ explicitly, multiply $U^\dagger U$, check you get I . **Both** steps are needed for full marks — computing $U^\dagger$ alone is an incomplete answer.

Ex. $T = \begin{pmatrix} \frac{1+i}{2} & 0 \\ 0 & \frac{2i}{\sqrt{4}} \end{pmatrix}$. Conjugating: $\overline{\frac{1+i}{2}} = \frac{1-i}{2}$ (the real part stays, only $i \rightarrow -i$), $\bar{i} = -i$. Then $T^\dagger T = \text{diag}\left(\frac{(1-i)(1+i)}{4}, 1\right) = \text{diag}\left(\frac{1}{2}, 1\right) \neq I \Rightarrow$ **not** unitary.

Trap: conjugating entries with a nonzero real part. $\overline{1+i} = 1-i$ — the flip applies to entries like $1+i$ exactly as it does to a pure imaginary $2i \rightarrow -2i$. This is the single most repeated error in this course; check every complex entry, not just the obviously imaginary ones.

Trap: stopping halfway. Question asks "show T is unitary" $\Rightarrow$ compute $T^\dagger$ and the product $T^\dagger T$ and state $= I$.

Trap: unitary vs. stochastic. Stochastic = real, non-negative, *columns* sum to 1. Unitary = complex, columns have *norm* 1 and are orthogonal. A matrix can be one and not the other.

Gate reference

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \sqrt{Z} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad \text{CNOT}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

CNOT: wire 1 controls, wire 2 flips. CNOT': wire 2 controls, wire 1 flips. Toffoli = CNOT with two controls. Rotations:

$$R_X(\theta) = e^{-iX\theta/2} = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_Y(\theta) =$$

$$\begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}, \quad R_Z(\theta) = \text{diag}(e^{-i\theta/2}, e^{i\theta/2}). \quad (\text{The } R_Y$$

entries above are copied from the lecture notes; computing $e^{-iY\theta/2}$ directly gives $+\sin \frac{\theta}{2}$ in the bottom-left, so the notes appear to have a sign typo there. Reproduce the notes' form if asked to "state" it; use $+\sin$ if asked to *derive* it.) Useful identities: $T^2 = S$, $S^2 = Z$, $Z^2 = I$, $XY = iZ$, $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, $H^2 = I$, $X|\pm\rangle = \pm|\pm\rangle$, $Z|\pm\rangle = |\mp\rangle$.

4. Quantum Fourier Transform / DFT [44.5

marks in practice e-tests]

Definition

$\text{DFT}_N := \frac{1}{\sqrt{N}}(\omega^{kl})_{k,l=0,\dots,N-1}$ with $\omega = e^{2i\pi/N}$ the N -th root of unity. $\omega^N = 1$, $\omega^i \neq 1$ for $0 < i < N$, $\omega^t = \omega^{t \bmod N}$. DFT_N is unitary.

The notes always say "DFT"; the exam taxonomy calls it "QFT". Same matrix. $N = 2$: $\omega = e^{i\pi} = -1$, so

$$\text{DFT}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H. \quad N = 4: \omega = i, \text{DFT}_4 =$$

$$\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}.$$

Period-extraction theorem (the examinable one)

If $t \mid N$ and $\psi_i = \sqrt{t/N}$ for $i = at + s$ (some a) and 0 otherwise — i.e. ψ is **t -periodic** — then $\phi = \text{DFT}_N \psi$ satisfies

$$|\phi_i| = \begin{cases} 1/\sqrt{t} & \text{if } \frac{N}{t} \mid i \\ 0 & \text{else.} \end{cases}$$

The first peak besides ϕ_0 sits at N/t . Period in $\Rightarrow$ N/period out (inverted). The shift s affects only phases, never $|\phi_i|$.

Circuit construction ($N = 2^n$)

$\text{DFT}_N |j\rangle = \bigotimes_{l=1}^n \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0.j_n-l+1 \dots j_n} |1\rangle)$, where $0.j_1 j_2 \dots$ is a **binary fraction** ($0.101_2 = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$). New gate: $R_k := \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$.

1. Load $|j_1\rangle$ on the top wire down to $|j_n\rangle$ on the bottom.
2. For each wire j_i top to bottom: apply H ; then for every wire j_k below it ($i < k \leq n$) apply R_{k-i+1} controlled by j_k .
3. Finish with a full SWAP reversing all wires (wire 1 $\leftrightarrow$ wire n , etc.).

$n = 3$: top wire gets H, R_2, R_3 ; middle gets H, R_2 ; bottom gets H ; then SWAP.

Trap: the wire that builds ψ_n . Counter-intuitively the *top* wire (j_1) builds the *last* tensor factor. That is exactly why the SWAP at the end is needed.

Trap: forgetting the SWAP. Without it you produce $\psi_n \otimes \dots \otimes \psi_1$ — reversed. (If you measure immediately afterwards, the SWAP may be done classically — cheaper — but say so.)

Trap: peak location. Output peaks at multiples of N/t , not of t .

Trap: "is $\text{DFT}_8 |110\rangle$ entangled?" **No.** The derivation above writes $\text{DFT}_N |j\rangle$ as an explicit tensor product of n single-qubit states, so $\text{DFT}_N |j\rangle$ is *never* entangled for a basis-state input. (DFT_N applied to a superposition certainly can be.)

5. Probability Distribution Vectors [33.5 marks in practice e-tests]

$d \in \mathbb{R}^N$ is a probability distribution $\iff d_i \geq 0 \forall i$ and $\sum_i d_i = 1$. Notation $\text{Pr}[x] = p$. Deterministic possibilities are always given a *fixed order* first.

Mean of the entries of an n -dimensional distribution vector: $\frac{1}{n} \sum_i x_i = \frac{1}{n}$ (the entries sum to 1 by definition).

Ex. Three fair coin tosses, count heads. Possibilities 0, 1, 2, 3 heads; $(\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8})^T$ (binomial $\binom{3}{k}/8$).

Ex. Die that never lands on 4, others equally likely. $(\frac{1}{5}, \frac{1}{5}, \frac{1}{5}, 0, \frac{1}{5}, \frac{1}{5})^T$ — the impossible outcome keeps its slot with entry 0; renormalise the rest over 5, not 6.

Trap: dropping the zero slot. The vector's dimension is the number of *possibilities*, not the number of *possible* outcomes. Keep the 0.

Trap: "non-negative", not "positive". $d_i \geq 0$ — entries may be 0. Say non-negative in a proof.

Trap: stating the ordering. Write down the ordering you're using before writing the vector; the vector is meaningless without it and the marks are often for stating it.

6. Quantum Measurement [30.5 marks in practice e-tests]

Complete measurement in the computational basis

For $\psi \in \mathbb{C}^N$: $\text{Pr}[\text{outcome} = x] = |\psi_x|^2$, and the **post-measurement state (p.m.s.)** is the basis vector $e_x = |x\rangle$.

Measuring changes the system. (Contrast: *observing* a probabilistic system is passive — it only updates knowledge, a third party can never tell you observed.)

$|a + bi|^2 = a^2 + b^2$. In particular $|i/\sqrt{2}|^2 = \frac{1}{2}$ and $|-3/10|^2 = 9/100$.

Ex. $\psi = (\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}})^T$. Amplitude of 0 = $\frac{1}{\sqrt{2}}$; $\text{Pr}[0] = \frac{1}{2}$. Amplitude of 1 = $\frac{i}{\sqrt{2}}$; $\text{Pr}[1] = \frac{1}{2}$. p.m.s. is $|0\rangle$ or $|1\rangle$ respectively.

Trap: answering only the numbers. If the question says "measure", it wants **(a) the probability of each outcome and (b) the post-measurement state**. Omitting the p.m.s. is a confirmed lost-marks pattern — re-read the question stem before moving on.

Trap: amplitude vs. probability. Amplitude can be complex/negative; probability is $|\cdot|^2$ and real. An "amplitude of 1" question wants ψ_1 itself, unsquared.

Trap: observe vs. measure. Use "observe" for probabilistic systems, "measure" for quantum. The notes deliberately separate the verbs and questions test the distinction.

7. Shor's Algorithm [30.5 marks in practice e-tests]

Three problems, in order

- **Factoring:** given $N = pq$ with distinct primes $p, q > 2$, find p, q .
- **Period finding:** given R and f with $f(x) = f(y) \iff x \equiv y \pmod{r}, r \leq R$, find r .
- **Order finding:** for a, N coprime, find the period r of $f(i) = a^i \pmod{N}$, i.e. smallest $r > 0$ with $a^r \equiv 1$. Write $r = \text{ord}_N a$.

Reduction factoring $\rightarrow$ order finding

1. Pick $a \in \{2, \dots, N-1\}$ coprime to N .

2. $r := \text{ord}_N a$ (with $R := N$, since $\text{ord}_N a \leq N$).
 3. If r is **odd**, restart at 1.
 4. $x := a^{r/2} + 1 \bmod N$, $y := a^{r/2} - 1 \bmod N$.
 5. If $\gcd(x, N) \in \{1, N\}$, restart at 1.
 6. Return $p = \gcd(x, N)$, $q = N / \gcd(x, N)$.
- Works because $xy = a^r - 1 \equiv 0 \pmod{N}$. If N is odd with ≥ 2 distinct prime factors, $\Pr[\text{restart}] \leq \frac{1}{2}$.

Quantum part (period finding)

Registers: n qubits top, m bottom, and $n \geq 2 \log R$.

$ \psi_1\rangle = 0^n\rangle \otimes 0^m\rangle$	start
$ \psi_2\rangle = 2^{-n/2} \sum_x x\rangle \otimes 0^m\rangle$	$H^{\otimes n}$ on top
$ \psi_3\rangle = 2^{-n/2} \sum_x x, f(x)\rangle$	U_f
$ \psi_4\rangle = \frac{\sqrt{r}}{\sqrt{2^n}} \sum_{x \equiv x_0(r)} x\rangle \otimes f(x_0)\rangle$	measure bottom
$ \psi_5\rangle = \text{DFT}_{2^n}(\text{top})$	peaks at multiples of $2^n/r$
$c = a \cdot 2^n/r$	measure top

If $r \mid 2^n$ you could just gcd several outputs c . In general $r \nmid 2^n$, so:

Post-processing (continued fractions)

With probability $\Omega(1/\log \log r)$ (in practice $\approx \frac{1}{3}$): $-\frac{r}{2} \leq rc \bmod 2^n \leq \frac{r}{2}$, hence $|\frac{c}{2^n} - \frac{d}{r}| \leq \frac{1}{2^{n+1}}$. **Theorem:** if $|\varphi - \frac{d}{r}| \leq \frac{1}{2L}$ and $r^2 \leq L$, then $\frac{d}{r}$ is a *convergent* of the continued fraction expansion of φ .

$$t = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}} = [a_0; a_1, a_2, \dots] \quad (a_i \text{ maximal})$$

Algorithm: with $\varphi = c/2^n$, $L = 2^n$, for each convergent γ : write $\gamma = d/r$; if $r \leq 2^n$ and d/r is $\frac{1}{2^{n+1}}$ -close to $c/2^n$, return r ; else next convergent.

Ex. $2.3 = 2 + \frac{1}{3 + \frac{1}{3}} = [2; 3, 3]$; convergents $[2] = 2$, $[2; 3] = 2\frac{1}{3}$. $0.99 = [0; 1, 99]$; convergents 0 and 1.

Trap: r odd or gcd trivial. Both send you back to step 1 — they are not failures of the theory.

Trap: simplified fraction. The convergent's denominator may not be the true r if d and r shared a factor. Small probability, already absorbed in the stated success probability — say this if asked why the algorithm can fail.

8. Partial Measurement [29 marks in practice e-tests]

Alternatives $A_1, \dots, A_m$ partition the possibility set ($A_i \cap A_j = \emptyset$, $\bigcup_i A_i = X$).

Quantum: $\Pr[\text{outcome} = k] = \sum_{x_i \in A_k} |\psi(x_i)|^2$.

p.m.s. in two steps: (1) $\phi_i^{(k)} = \psi_i$ if $x_i \in A_k$, else 0; (2) normalise: $\text{p.m.s.} = \frac{\phi^{(k)}}{\|\phi^{(k)}\|} = \frac{\phi^{(k)}}{\sqrt{\Pr[\text{outcome} = k]}}$.

Probabilistic: $\Pr[k] = \sum_{i \in A_k} \mu_i$; conditional $v_i = \mu_i / \Pr[k]$ if $i \in A_k$, else 0. *No division by a square root* — probabilities, not amplitudes.

This is a projective measurement in the computational basis. Other kinds exist (projective in another basis, generalized, POVM) — the course means this one.

Ex. $\psi = (\frac{1}{10}, -\frac{3}{10}, \frac{9}{10}i, \frac{3}{10})^T$ over paths (left, right, top, bottom). $A_{\text{hor}} = \{\text{left, right}\}$, $A_{\text{ver}} = \{\text{top, bottom}\}$. $\Pr[\text{hor}] = \frac{1}{100} + \frac{9}{100} = \frac{1}{10}$, $\Pr[\text{ver}] = \frac{81}{100} + \frac{9}{100} = \frac{9}{10}$.
p.m.s. (hor) = $(\frac{1}{10}, -\frac{3}{10}, 0, 0)^T / \sqrt{\frac{1}{10}} = (\frac{1}{\sqrt{10}}, -\frac{3}{\sqrt{10}}, 0, 0)^T$.

Composite measurements — building the C sets

$M_A \otimes M_B$ on AB has alternatives $C_{ij} = A_i \times B_j$ (*Cartesian product*: concatenate one symbol from A_i with one from B_j). Measuring only A : $M_A \otimes I$, alternatives $C_i = A_i \times \{y_1, \dots, y_M\}$ — i.e. the B slot is a wildcard. For three systems with $M_1 \otimes I \otimes M_2$: a string $s_1 s_2 s_3 \in C_{jk} \iff s_1 \in A_j$ and $s_3 \in B_k$; the middle symbol is unconstrained.

Ex. M_1 on $\{x, y, z\}$ with $A_0 = \{x, y\}$, $A_1 = \{z\}$; M_2 likewise. Then $xyy \in C_{00}$ (first $y \in A_0$, last $y \in A_0$), and $xyy \notin C_{10}$. $zzz \in C_{11}$; $zyz \in C_{11}$; $xyz \in C_{01}$.

Trap: normalising by $\Pr$ instead of $\sqrt{\Pr}$ in the quantum case (that's the *probabilistic* rule).

Trap: dropping the zeros. The p.m.s. lives in the *same* space as ψ — keep all N entries, with zeros outside the alternative.

Trap: global phase. Partial-then-full measurement and direct full measurement can give p.m.s. differing by c with $|c| = 1$. State that this is a global phase and physically identical — do not "fix" it.

9. Quantum Teleportation [12 marks in practice e-tests]

Alice holds $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ (wire A1) and shares $|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ (wires A2, B) with Bob. Only classical communication allowed. Circuit: CNOT (A1→A2), H on A1, measure A1→ a and A2→ b , send a, b ; Bob applies X if $b = 1$, then Z if $a = 1$.

The six states — this is the whole question

$$\begin{aligned} |\phi_1\rangle &= |\psi\rangle \otimes |\beta_{00}\rangle = \frac{\alpha}{\sqrt{2}}(|000\rangle + |011\rangle) + \frac{\beta}{\sqrt{2}}(|100\rangle + |111\rangle) \\ |\phi_2\rangle &= (\text{CNOT} \otimes I_2) |\phi_1\rangle \\ &= \frac{\alpha}{\sqrt{2}}(|000\rangle + |011\rangle) + \frac{\beta}{\sqrt{2}}(|110\rangle + |101\rangle) \\ |\phi_3\rangle &= (H \otimes I_4) |\phi_2\rangle = \frac{\alpha}{2}(|000\rangle + |100\rangle + |011\rangle + |111\rangle) \\ &\quad + \frac{\beta}{2}(|001\rangle - |101\rangle + |010\rangle - |110\rangle) \\ |\phi_4\rangle &= \text{p.m.s. after measuring } ab \quad (\Pr = \frac{1}{4} \text{ for each } ab) \\ |\phi_5\rangle &= (I_4 \otimes X)^b |\phi_4\rangle \\ |\phi_6\rangle &= (I_4 \otimes Z)^a |\phi_5\rangle = |ab\rangle \otimes |\psi\rangle \end{aligned}$$

Read ϕ_4 straight off ϕ_3

Collect the two terms of $|\phi_3\rangle$ whose first two qubits are ab , then divide by $\sqrt{\Pr} = \sqrt{1/4} = \frac{1}{2}$ (i.e. multiply by 2):

ab	$ \phi_4\rangle$	Bob's qubit	correction
00	$ 00\rangle \otimes (\alpha 0\rangle + \beta 1\rangle)$	ψ	none
01	$ 01\rangle \otimes (\alpha 1\rangle + \beta 0\rangle)$	$X\psi$	X
10	$ 10\rangle \otimes (\alpha 0\rangle - \beta 1\rangle)$	$Z\psi$	Z
11	$ 11\rangle \otimes (\alpha 1\rangle - \beta 0\rangle)$	$XZ\psi$	X then Z

$\Pr[ab] = \left|\frac{\alpha}{2}\right|^2 + \left|\frac{\beta}{2}\right|^2 = \frac{|\alpha|^2 + |\beta|^2}{4} = \frac{1}{4}$ for every ab — independent of ψ , since $|\alpha|^2 + |\beta|^2 = 1$. Quote that one-liner; it is usually worth its own marks.

Ex. $\psi = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)^T$ ($\alpha = \frac{1}{\sqrt{2}}, \beta = -\frac{1}{\sqrt{2}}$), outcome $ab = 11$.
 $|\phi_4\rangle = |11\rangle \otimes \left(\frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|0\rangle\right) = |11\rangle \otimes |+\rangle$.
 $|\phi_5\rangle = (I_4 \otimes X)|\phi_4\rangle = |11\rangle \otimes |+\rangle$ (since $X|+\rangle = |+\rangle$).
 $|\phi_6\rangle = (I_4 \otimes Z)|\phi_5\rangle = |11\rangle \otimes |-\rangle = |11\rangle \otimes |\psi\rangle$ ✓
 $\Pr[11] = \frac{1}{4}$.

Trap: correction order. X first (controlled by b , the second bit), then Z (controlled by a). Getting a and b the wrong way round flips two of the four cases.

Trap: leaving out Alice's two qubits. $|\phi_4\rangle$ – $|\phi_6\rangle$ are still 3-qubit states — write $|ab\rangle \otimes (\dots)$, not just Bob's qubit, unless the question asks only for Bob's.

Trap: no cloning. Alice's original qubit is destroyed by the measurement; teleportation moves, not copies.

10. Stabilizer Formalism [26 marks in practice e-tests]

ψ is **stabilized** by a set M of unitaries $\iff U\psi = \psi$ for all $U \in M$.

Pauli = all n -qubit tensor products of $\{X, Y, Z, I\}$ with factors $\pm 1, \pm i$.

ψ is a **stabilizer state** $\iff \exists M \subseteq \text{Pauli}$ stabilizing ψ and *only* ψ (up to global phase). Notation $\psi \sim M$.

Update rule: if $\psi \sim M$ then $U\psi \sim M^U := \{UVU^\dagger : V \in M\}$.

For any Clifford gate G ($\{\text{CNOT}, H, S\}$), $M \subseteq \text{Pauli} \implies M^G \subseteq \text{Pauli}$.

Single-qubit dictionary

$\{Z\}$	$ 0\rangle$	$\{-Z\}$	$ 1\rangle$
$\{X\}$	$ +\rangle$	$\{-X\}$	$ -\rangle$
$\{Y\}$	$\frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$\{-Y\}$	$\frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$

Multi-qubit: $\{Z \otimes I, I \otimes Z\} \rightarrow |00\rangle$; $\{Z \otimes I, -I \otimes Z\} \rightarrow |01\rangle$; $\{X \otimes I, I \otimes Z\} \rightarrow |+\rangle \otimes |0\rangle$; $\{Z \otimes Z, X \otimes X\} \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$; $|0\rangle^{\otimes n} \sim \{Z \otimes I \otimes \dots \otimes I, I \otimes Z \otimes \dots \otimes I, \dots, I \otimes \dots \otimes I \otimes Z\}$ (n generators). $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) \sim \{Z \otimes Z \otimes I, I \otimes Z \otimes Z, X \otimes X \otimes X\}$.

Conjugation table (how Cliffords move Paulis)

	$X \mapsto$	$Y \mapsto$	$Z \mapsto$
H	Z	$-Y$	X
S	Y	$-X$	Z

CNOT (control c , target t): $X_c \mapsto X_c X_t$, $Z_c \mapsto Z_c$, $X_t \mapsto X_t$, $Z_t \mapsto Z_c Z_t$. Signs carry through: $-Y \otimes Z$ conjugated by $H \otimes I$ gives $-(HYH) \otimes Z = +Y \otimes Z$.

Gottesman–Knill

A circuit using only Clifford gates can be simulated efficiently classically. Simulation: $M_0 = \{Z \otimes I \otimes \dots\}$ for $|0\rangle^{\otimes n}$, then $M_i = M_{i-1}^{G_i}$ for each gate, then linear algebra for the measurement outcomes. $|M_i| = n$ always, so it's polynomial. **Consequence: Clifford gates alone give no quantum speedup.**

Trap: forgetting the $\pm$. $\{X\}$ and $\{-X\}$ stabilize *different* states ($|+\rangle$ vs $|-\rangle$). Carry every sign through the conjugation.

Trap: $UVU^\dagger$, not $U^\dagger VU$. Order matters.

11. Grover's Algorithm [20 marks in practice e-tests]

Ingredients

$|*\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$ (uniform superposition).

$V_f |x\rangle = \begin{cases} -|x\rangle & f(x) = 1 \\ |x\rangle & \text{else} \end{cases}$ built from U_f with the ancilla in $|-\rangle$ (phase kickback); the ancilla stays $|-\rangle$ and is discarded.

$\text{FLIP}_0 |x\rangle = \begin{cases} |0\rangle & x = 0 \\ -|x\rangle & \text{else} \end{cases}$ — a Z with negative controls on all other wires.

$\text{FLIP}_* |\psi\rangle = \begin{cases} |*\rangle & |\psi\rangle = |*\rangle \\ -|\psi\rangle & \langle \psi | * \rangle = 0 \end{cases} = H^{\otimes n} \text{FLIP}_0 H^{\otimes n}$.

Algorithm

1. Start $|0^n\rangle$; apply $H^{\otimes n}$ to get $|*\rangle$.
2. Apply V_f , then FLIP_* .
3. Repeat step 2 t times.
4. Measure $\Rightarrow x_0$ with high probability.

Why it works (the geometry)

Split $|*\rangle = \frac{1}{\sqrt{2^n}} |x_0\rangle_{\text{good}} + \sqrt{\frac{2^n-1}{2^n}} (\text{bad})$. $\cos \theta = |\langle * | \text{bad} \rangle| = \frac{\sqrt{2^n-1}}{\sqrt{2^n}}$, so $\theta \approx \frac{1}{\sqrt{2^n}}$. V_f reflects about the *bad* axis; FLIP_* reflects about $|*\rangle$. **Two reflections = a rotation by 2θ toward good.**

$$t \approx \frac{\pi/2}{\theta} - \frac{1}{2} \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4} \sqrt{2^n} \Rightarrow O(\sqrt{2^n}) \text{ queries.}$$

Trap: more iterations $\neq$ better. The rotation overshoots past $\pi/2$ and the success probability falls again. t is fixed by the formula.

Trap: quadratic, not exponential. Classical $\Theta(2^n) \rightarrow$ quantum $\Theta(2^{n/2})$, and Theorem 19.1 says $\Omega(2^{n/2})$ queries are *necessary* — Grover is optimal for black-box SAT.

12. Quantum Error Correction [20 marks in practice e-tests]

Two obstacles vs. classical repetition

- (1) Majority voting would require measuring the data, destroying superposition — so measure a *syndrome* instead.
- (2) A qubit has continuously many possible errors, not just a flip.

Bit-flip code (corrects X)

$|0\rangle \mapsto |000\rangle$, $|1\rangle \mapsto |111\rangle$; encode with two CNOTs from the data wire. Note $|+\rangle \mapsto \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$, *not* $|+++ \rangle$. Syndrome via two ancillas: $m_1 = a \oplus b$, $m_2 = a \oplus c$.

$m_1 m_2 = 00$: no error 01: wire 3 10: wire 2 11: wire 1

Then apply X to that wire.

Phase-flip code (corrects Z)

Same structure conjugated by H : $|0\rangle \mapsto |+++ \rangle$, $|1\rangle \mapsto |-- \rangle$; syndrome circuit is the bit-flip one with H on each data wire before and after.

Shor's code (9 qubits, corrects arbitrary 1-qubit error)

Phase-flip code on the outside, bit-flip code inside each of the three blocks:

$$|\tilde{0}\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)^{\otimes 3}, \quad |\tilde{1}\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)^{\otimes 3}.$$

X -error: treat each block as its own bit-flip code (compare the three qubits *within* a block). Z -error: shows up as a *relative sign flip between blocks* — a block reading $(|000\rangle - |111\rangle)$ where the others read $+$ means apply Z to any one wire of that block. Y = both. **Theorem:** a code that corrects single X , Y , Z errors (by measure-then-correct) corrects *arbitrary* single-qubit errors. Hence Shor's code corrects any 1-qubit error.

Steane code (7 qubits, corrects 1 error)

$$|\tilde{0}\rangle = \frac{1}{\sqrt{8}}(|0000000\rangle + |1010101\rangle + |0110011\rangle + |1100110\rangle + |0001111\rangle + |1011010\rangle + |0111100\rangle + |1101001\rangle)$$

$|\tilde{1}\rangle$ = the same eight strings bit-flipped. Preferred for fault tolerance: more gates are transversal.

Trap: measuring the data. The whole construction only works because m_1, m_2 are read from *ancillas*. Measuring the three data wires destroys the superposition being protected.

Trap: $\tilde{X}$ vs $\tilde{Z}$ on Shor's code. Logical $\tilde{X}$ is implemented by Z on qubits 1, 4, 7 (or $Z^{\otimes 9}$); logical $\tilde{Z}$ by $X^{\otimes 9}$. The letters swap — a favourite exam item.

13. Bernstein–Vazirani Algorithm [17.5 marks in practice e-tests]

Secret $s \in \{0, 1\}^n$, $f(x) = x \cdot s = \sum_i x_i s_i \bmod 2$. Classical: n queries (with $m \leq n$ queries, success probability 2^{m-n}). Quantum: **one**.

The four states — memorise this ladder

$$\begin{aligned} |\psi_0\rangle &= |0\rangle^{\otimes n} \otimes |1\rangle \\ |\psi_1\rangle &= (H^{\otimes n} \otimes H) |\psi_0\rangle = |+\rangle^{\otimes n} \otimes |-\rangle \\ &= \frac{1}{\sqrt{2^n}} \sum_x |x\rangle \otimes |-\rangle \\ |\psi_2\rangle &= U_f |\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle \otimes |-\rangle \\ |\psi_3\rangle &= (H^{\otimes n} \otimes I) |\psi_2\rangle = |s\rangle \otimes |-\rangle \end{aligned}$$

$U_f |x, y\rangle = |x, y \oplus f(x)\rangle$. **Phase kickback:** with the ancilla in $|-\rangle$, $U_f(|x\rangle \otimes |-\rangle) = (-1)^{f(x)} |x\rangle \otimes |-\rangle$ — the output

of f lands as a sign on the *input* register and the ancilla is unchanged.

Circuit-completion questions (the e-test format)

Match the target state to the ladder and name the missing box:

Target outcome	Fill in
$\sum 2^{-n/2} x, f(x)\rangle$	$H^{\otimes n}$ then U_f , ancilla $ 0\rangle$
$\sum 2^{-n/2} (-1)^{f(x)} x\rangle \otimes -\rangle$	$H^{\otimes n} \otimes H$ on $ 0^n 1\rangle$, then U_f
$\sum 2^{-n/2} x, f(0^n)\rangle$	constant f : U_f acts as a fixed offset
$\sum 2^{-n/2} x\rangle \otimes 0\rangle$	apply U_f twice (it is its own inverse)

Constant / balanced special cases

$f \equiv 0$: every $(-1)^{f(x)} = +1$, state is $|+\rangle^{\otimes n}$, $H^{\otimes n}$ returns $|0^n\rangle \Rightarrow$ measurement gives 00000. $f \equiv 1$: global -1 only $\Rightarrow$ also 00000 (global phase invisible). **Balanced** f : half the x give $(-1)^{f(x)} = +1$, half give -1 , so $\sum_x (-1)^{f(x)} = 0$, so the amplitude of $|0^n\rangle$ is 0 and $\Pr[0^n] = 0$ — the outcome is *never* 0^n .

Trap: ancilla state. $|-\rangle$ (from $|1\rangle$ then H), not $|+\rangle$. With $|+\rangle$ there is no kickback.

Trap: good/bad replacements for U_f . A replacement is *good* iff it is unitary and produces the same $(-1)^{f(x)}$ phase pattern; e.g. $|x, y\rangle \mapsto |x, y \oplus f(x)\rangle$ is good, $|x, y\rangle \mapsto |x, f(x)\rangle$ is *not* (not invertible).

14. Quantum Amplitudes and Probabilities

[17 marks in practice e-tests]

$\psi \in \mathbb{C}^N$ with $\sum_i |\psi_i|^2 = 1$. Entry $\psi_x =$ **amplitude** of x (may be negative or complex). $\Pr[x] = |\psi_x|^2$.
 $|x\rangle := e_x$; $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.
 $\langle \psi | := |\psi\rangle^\dagger$; $\langle \psi | \phi \rangle = \langle \psi | \cdot | \phi \rangle$ (inner product).

Applying a gate: $\psi' = U\psi$, then re-read amplitudes.

Ex. $H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)^T$, so $\Pr[0] = \Pr[1] = \frac{1}{2}$.

Bra / bracket

$\langle \psi | = |\psi\rangle^\dagger$ — a *row* vector with every entry conjugated.

Ex. $\psi = \begin{pmatrix} 0 \\ i \end{pmatrix} \Rightarrow \langle \psi | = (0 \quad -i)$. Then with $\phi = \left(\frac{i}{\sqrt{2}}, -\frac{i}{\sqrt{2}}\right)^T$:
 $\langle \psi | \phi \rangle = 0 \cdot \frac{i}{\sqrt{2}} + (-i) \left(-\frac{i}{\sqrt{2}}\right) = -\frac{1}{\sqrt{2}}$. Note $i \cdot i = -1$, so the sign flips — a standard slip.

Trap: $|\cdot|^2$ on complex entries. $|a + bi|^2 = a^2 + b^2$, *not* $(a + bi)^2$.

Trap: conjugating the bra. $\langle \psi | \phi \rangle$ conjugates the *left* argument only.

Trap: negative amplitudes are legal, negative probabilities are not. Amplitudes interfere and cancel; that cancellation is where every speedup comes from.

15. Probabilistic Matrices [14 marks in practice e-tests]

$A \in \mathbb{R}^{N \times N}$ is a probabilistic process (stochastic matrix) $\iff$ **every column** is a valid probability distribution: all entries ≥ 0 and **each column sums to 1**. Applying it: $y = Ax$.

Column j = "where the system goes, given it is currently in state j ". Reading a word problem: for each source state, write its outgoing probabilities *down* a column.

Ex. Weather, order (Rain, Sun, Cloud). "From Rain: 50% rain, 20% sun, 30% cloud" fills column 1 as $(0.5, 0.2, 0.3)^T$. Repeat for each source. Then verify all three columns sum to 1.

Trap: rows vs. columns. Rows summing to 1 is the wrong test and is a deliberately planted distractor.

Trap: stating the test but not applying it. If asked "is P valid?", state *both* conditions **and** check them against the given numbers, naming the offending column if it fails.

Trap: blank-entry questions: solve via the column constraint (entry = 1 – the rest of its column).

16. Physical Qubit Implementation [12 marks in practice e-tests]

Ion trap: the system

Ions trapped between two vibrating electromagnetic fields. Total state space $\mathbb{C}_{\text{vibration}}^3 \otimes \mathbb{C}_{\text{ion}_1}^2 \otimes \dots \otimes \mathbb{C}_{\text{ion}_n}^2$ (vibration levels $|0\rangle, |1\rangle, |2\rangle$; each ion $|0\rangle, |1\rangle$).

The three required operations

1. Initialize ("cooling")

- Doppler cooling:** laser slightly *below* the transition energy. When the ion moves *towards* the laser, the Doppler effect raises the perceived frequency/energy, so absorption is likelier — photons push back on the swing and damp the vibration.
- Sideband cooling:** shine energies $E_{\text{big}} - E_1$ and $E_{\text{big}} - E_2$ but **not** $E_{\text{big}} - E_0$. Electrons in E_1, E_2 get pumped up and fall back randomly; those landing in E_0 are never re-excited, so population accumulates in $|0\rangle$.

2. Measure: laser at $E_{\text{aux}} - E_0$. If the ion is $|0\rangle$ it absorbs, re-emits, and **lights up**; if $|1\rangle$ nothing is absorbed and it stays dark.

3. Unitaries: laser at angular frequency $\omega + \varphi$ makes $H_\varphi = C \begin{pmatrix} 0 & e^{-i\varphi} \\ e^{i\varphi} & 0 \end{pmatrix}$, and $U_{\varphi,t} = e^{-iH_\varphi t}$. E.g. $U_{\pi,\pi/2} = X$; $U_{\pi/2,\pi/2}$ then $U_{\pi,\pi/2}$ gives Z ; $U_{\pi/2,\pi/4}$ then $U_{\pi,\pi/2}$ gives H . Multi-qubit gates use the shared vibration mode (the gap $\Delta E = E_{20} - E_{11}$).

17. Bomb Tester (Elitzur–Vaidman) [6 marks in practice e-tests]

Beam splitter $B = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (= Hadamard), states (up, down).

	no bomb	bomb
up	1	1/4
down	0	1/4
explodes	0	1/2

No bomb: $B(\frac{1}{0}) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$, then B again = $(\frac{1}{0})$ — interference sends it to *up* with certainty.

Bomb: the bomb *measures* which path was taken. $\Pr[\text{down}] = \Pr[\text{up}] = \frac{1}{2}$; down $\Rightarrow$ explosion. Up $\Rightarrow$ p.m.s.

$(\frac{1}{0})$, second splitter gives $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^T$, so $\frac{1}{4}$ each.

The punchline: measuring *down* at the end can only happen if a bomb is present — so with probability $\frac{1}{4}$ you detect the bomb *without the photon ever entering the box*. Impossible classically. An improved setup pushes this to ≈ 1 with ≈ 0 explosion probability.

Trap: why the two cases differ. Nothing hits the photon in the bomb case either — the difference is that the bomb *destroys the superposition* (it is a measurement), so no interference happens at the second splitter.

18. Quantum State Validity [4 marks in practice e-tests]

$\psi \in \mathbb{C}^N$ is a valid quantum state $\iff \sum_i |\psi_i|^2 = 1$. No non-negativity requirement — entries may be negative, imaginary, or complex.

Ex. $(\frac{1}{\sqrt{2}}, \frac{i}{\sqrt{2}})$: $\frac{1}{2} + \frac{1}{2} = 1 \checkmark$. $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$: $4 \cdot \frac{1}{4} = 1 \checkmark$. $(\frac{1}{2}, \frac{1}{2})$: $\frac{1}{2} \neq 1 \times$.

Trap: summing entries instead of squared moduli — that's the *probabilistic* test.

19. Quantum Gate Universality [0 so far — but cheap marks in practice e-tests]

G is **universal** $\iff$ any unitary can be approximated to arbitrary precision by circuits over G (no auxiliary qubits). "Universal for single-qubit gates" = same, restricted to $n = 1$.

Set	Universal?
Pauli $\{X, Y, Z\}$	No — and not even for 1 qubit. Products of Paulis are Paulis up to phase; they can never entangle.
$\{\text{CNOT, Toffoli, } X, \text{SWAP}\}$	No — each maps basis states to basis states, so no superposition and no speedup.
$\{R_X, R_Y, R_Z\}$	Universal <i>for single-qubit</i> gates (two of the three suffice).
$\{\text{all single-qubit gates, CNOT}\}$	Yes
$\{R_X, R_Y, R_Z, \text{CNOT}\}$	Yes
Clifford $\{\text{CNOT, } H, S\}$	No (Gottesman–Knill: classically simulable)
$\{\text{Clifford, } T\}$	Yes
$\{\text{Toffoli, } H\}$	Yes

Why Paulis fail, in one line: $XY = iZ$, so you get Z only

up to global phase, and every product stays in $\{I, X, Y, Z\}$ up to phase — a finite set, single-qubit only.

20. Fault-Tolerant Computation

Transversal gates

A gate is **transversal** if it can be implemented by applying identical gates to each physical qubit separately. On Shor's code: $\tilde{X} = Z$ on qubits 1,4,7 (or $Z^{\otimes 9}$), $\tilde{Z} = X^{\otimes 9}$; but $\tilde{H}$ is *not* available this simply. On the Steane code: $\tilde{G} = G^{\otimes 7}$ for $G \in \{X, Y, Z\}$; $\tilde{S} = S^{\otimes 7}$ followed by $Z^{\otimes 7}$; $\widetilde{\text{CNOT}} =$ seven CNOTs. $\tilde{T}$ is **not** transversal but is implementable by a more complex method — needed, since Clifford alone isn't universal.

The three FT requirements

- (A) FT gate: ≤ 1 physical error *in the inputs* $\Rightarrow \leq 1$ error per output block.
 (B) FT gate: no input error and ≤ 1 physical *gate* error $\Rightarrow \leq 1$ error per output block.
 (C) Error correction: no input error and ≤ 1 error occurred $\Rightarrow \leq 1$ error in the output.
 Transversal gates satisfy (A) and (B) automatically.

Construction: encode every qubit as a block; replace every gate by its FT implementation; insert an error correction step *after each gate*.

The error-probability calculation

p = physical gate error probability, G = number of physical gates per FT gate, errors independent.

$$P'_1 \leq Gp + P_1, \quad P'_2 \leq (Gp)^2 + P_1Gp + P_2, \quad P''_1 \leq Gp$$

$$P''_2 \leq 5(Gp)^2 + P_2 \quad (\text{assuming } P_1 \leq Gp)$$

So each logical gate adds $\leq 5(Gp)^2$; over l logical gates, $P_2 \leq 5l(Gp)^2$. **The error shrinks iff** $5(Gp)^2 < p$, i.e. $p < \frac{1}{5G^2}$ — this is the threshold.

Ex. $G = 50$. $p = \frac{1}{10000}$: $5(Gp)^2 = \frac{5 \cdot 2500}{10^8} = \frac{1}{8000} > p \Rightarrow$ fails.
 $p = \frac{1}{15000}$: $5(Gp)^2 = \frac{5 \cdot 2500}{2.25 \cdot 10^8} = \frac{1}{18000} < p \Rightarrow$ works.

Threshold theorem: if the physical error rate is below the threshold, recursion (one more FT layer per level) drives the logical error arbitrarily low. Each layer multiplies the gate count by G , so k layers cost G^k physical gates per logical gate. Real-world threshold believed to be between **0.1% and 1%**.

Recursion sizing (exam pattern)

Error after k layers: iterate $p_{i+1} = 5(Gp_i)^2$. Equivalently, with $q_i := 5G^2p_i$, $q_{i+1} = q_i^2$, so $q_k = q_0^{2^k}$ — the exponent

doubles per layer. To reach 10^{-50} from $p = 10^{-4}$ you need the smallest k with $q_0^{2^k} \leq 5G^2 \cdot 10^{-50}$.

21. Quantum Complexity

NP: $L \in \text{NP}$ iff there is a poly-time classical A with — for YES-instances x , some witness ω has $A(x, \omega) = 1$; for NO-instances, no ω does. SAT (does some z have $f(z) = 1$?) is NP-complete.

Oracle lower bound: classically $\Theta(2^n)$ queries to distinguish YES/NO for black-box SAT; quantumly $\Theta(2^{n/2})$. Achieved by Grover; and **Theorem 19.1:** any quantum algorithm distinguishing them with constant error needs $\Omega(2^{n/2})$ queries to V_f . So no black-box quantum algorithm solves SAT in polynomial time.

QMA: $L \in \text{QMA}$ iff a poly-time *quantum* verifier exists with — YES: some quantum witness $|\psi\rangle$ has $\Pr[A(x, |\psi\rangle) = 1] \geq \frac{2}{3}$; NO: *all* $|\psi\rangle$ give $\Pr \leq \frac{1}{3}$. The quantum analogue of NP.

2-local Hamiltonian: $H = \sum_{i=1}^k H_i$ with each H_i acting on only two qubits. Deciding whether the smallest eigenvalue is $\leq \frac{1}{3}$ (YES) or $\geq \frac{2}{3}$ (NO) is **QMA-complete**.

22. Cross-cutting reminders

Say these out loud before the exam

- 1. Finish the question.** Both failed reviews in this course were right-numbers-wrong-completeness: tensor order reversed, and post-measurement state omitted. After you get the number, re-read the stem and check every sub-part is answered.
- 2. Conjugate means conjugate.** $\overline{1+i} = 1-i$. Not just the pure imaginaries.
- 3. Columns, not rows,** for stochastic matrices.
- 4. Global phase c , $|c| = 1$:** name it, call the states physically equal, move on. The notes do this every single time it appears.
- 5. State your ordering** of possibilities before writing any vector.
- 6. Two eigenvalues** for a 2×2 ; the zero vector is never an eigenvector.
- 7. Measuring changes a quantum system; observing a probabilistic one does not.**

Notation

$:=$ definition $\dagger$ conjugate transpose $\otimes$ tensor product $H^{\otimes n}$ H on all n wires $\tilde{X}, |\tilde{0}\rangle$ logical (encoded) operator/qubit $0.j_1j_2$ binary fraction p.m.s. post-measurement state.